AMD AI DEVMASTER HACKATHON · TRACK 1 · MULTIMODAL AI

Katha





Speak a story idea. Get back a storybook — written, illustrated, narrated, and bound into a video. Every model local on an AMD Radeon GPU.

N DIVIJ · solo developer

github.com/N-45div/Katha

Radeon PRO W7900 · ROCm 7.2.1

THE PROBLEM

“One more story?”



Prompt boxes exclude kids

Story tools are typed-prompt tools. **A five-year-old can't type a prompt** — but they can *say* one, vividly, in the language they dream in. If the input is a keyboard, the child is a spectator.



The storytelling internet speaks English

A child in Jaipur gets stories in a language their grandmother doesn't tell stories in. **Story text and voice should follow the child's language** — not the training data's default.



Children's voices, uploaded

Bedtime tools record children speaking and invent stories about them — then send it all to a cloud API. **That data should never leave the house.**

THE PRODUCT

Speak it. Watch it become a book.

A concrete session

A parent taps the mic; their daughter says **“एक हाथी जो उड़ना चाहता था”** — an elephant who wanted to fly. Katha detects Hindi, writes a six-page story with a named hero, paints the hero once, holds that same elephant across every page, narrates it in a Hindi voice, and binds a subtitled video with gentle pan-and-zoom — plus a printable A4 PDF.

Page four not right? **One click repaints just that page.**

Every run produces

- 🎬 **storybook.mp4** — narrated, subtitled, Ken Burns motion
- 📖 **storybook.pdf** — print-ready, Devanagari/CJK-correct
- 🖼️ Page gallery + per-page repaint
- 🗣️ 8 languages, auto-detected from voice
- 🎨 3 art styles · reproducible seeds

REAL OUTPUT — FIRST VALIDATED RUN ON THE RADEON PRO W7900

“The Brave Dragon”







Frames from `storybook.mp4` — one spoken idea (*“a dragon who is afraid of fire”*), one command, 2.6 minutes on the GPU. Same hero on every page; subtitles and narration produced by the pipeline itself.

HOW IT WORKS

Five models, one Radeon, zero cloud

INPUT

- 🎙️ **Voice note** — 8 languages, auto-detected
- ⌨️ **...or a typed idea** — optional path

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AMD RADEON GPU · ROCM — SEQUENTIAL LOAD/UNLOAD, ONE MODEL RESIDENT AT A TIME

- **Whisper turbo** — speech → idea + language
- **Qwen2.5-7B** — validated JSON story script
- **SDXL + IP-Adapter** — hero portrait → consistent pages
- **Kokoro-82M** — narration in your language

▼

OUTPUT

- **ffmpeg** — motion · subtitles · assembly
- 🎬 **storybook.mp4** — the bedtime deliverable
- 📖 **storybook.pdf** — the bookshelf deliverable

RADEON PERFORMANCE — MEASURED, NOT ESTIMATED

2.6 minutes from idea to storybook

2.6 min
4-PAGE BOOK, END-TO-END

~7 s
PER 1024² ILLUSTRATION, STEADY-STATE

14.3 GB
PEAK VRAM — FITS 16 GB RADEONS

0
CLOUD API CALLS

STAGE	TIME	PEAK VRAM	NOTE
story — Qwen2.5-7B fp16	30.5 s	14.3 GB	structured script, one-retry JSON contract
illustrate — SDXL ×5 @ 30 steps	60.5 s	9.7 GB	4.07 it/s after one-time kernel warm-up
narrate — Kokoro-82M	48.4 s	0.8 GB	title + every page, 24 kHz
assemble — ffmpeg	17.4 s	—	CPU encode

Radeon PRO W7900 (gfx1100, 48 GB) · ROCm 7.2.1 · PyTorch 2.9.1 · fp16 throughout · every run writes its own `benchmark.json`

THE CRAFT

What makes it a book, not four pictures

The hero stays the hero

Katha paints a **reference portrait first**, then conditions every page on it with IP-Adapter (scale 0.55), layered over a shared character sheet and deterministic per-page seeds. If adapter weights are missing, it **falls back gracefully** — the pipeline cannot be killed by a missing optimization.

Prompts as a budget

SDXL's CLIP truncates at 77 tokens — we caught it silently chopping style suffixes on-GPU. Now style presets are compact and the character sheet and scene are **word-capped in code**; composed prompts sit at ~48 words with room to spare.

One multilingual rule

Narration in the speaker's language; scene prompts always in English. Whisper detects, Qwen writes both, Kokoro speaks it, and the PDF exporter picks Devanagari/CJK-capable fonts. Unsupported languages degrade to English with a warning — never a crash.

PRACTICAL VALUE

Who picks this up tomorrow



Families

The nightly story, personalised, in minutes — driven by the child's own voice. Private by architecture: **nothing leaves the GPU.**



Regional-language classrooms

Reading material in the language of instruction, on school hardware, offline — with a **printable PDF** for every desk.



Creators & kiosks

Subtitled story videos for kids' channels outside English; library story-corners on one Radeon workstation. **No per-use cost, no accounts.**

STATUS & ASK

Validated on Radeon. Ready to grow.

Today

- Five-stage pipeline **validated end-to-end on a W7900**, benchmarks measured on-GPU
- Web app (mic → book) + scriptable CLI, MIT-licensed
- Two-command setup on any ROCm box; region-hardened installers

Next

- Multi-chapter heroes · on-device child-safety filter
- Tamil, Telugu, Bengali narration
- Quantized variants: under 8 GB for entry Radeons & Ryzen AI

Katha turns a child's spoken idea into their own storybook in under three minutes — private, multilingual, and entirely on an AMD Radeon GPU.

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